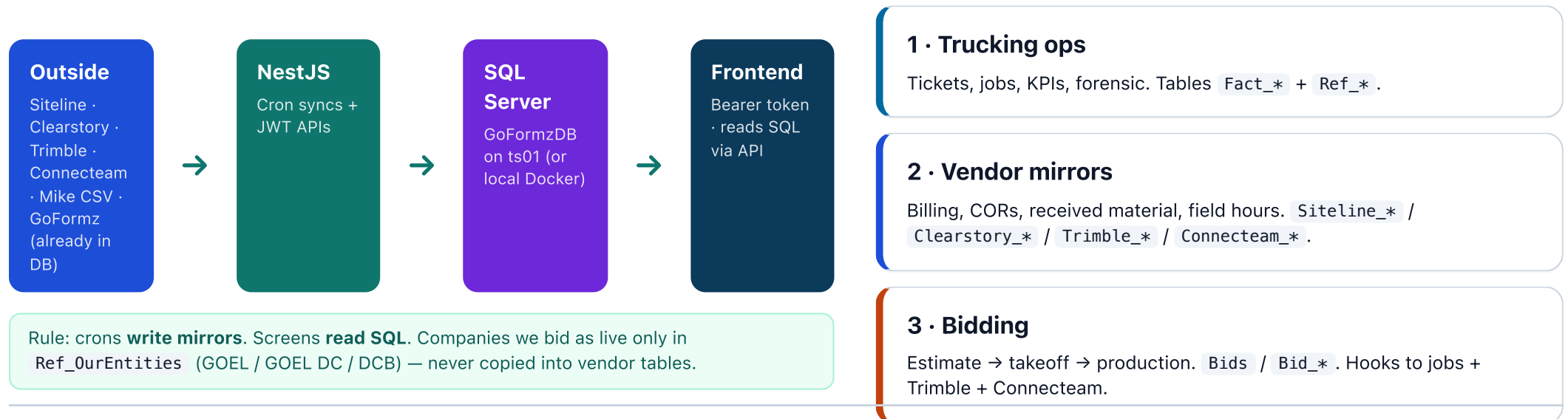


Backend design

One NestJS API in front of SQL Server `GoFormzDB`. Vendors sync into tables. The UI never talks to Sitrine / Clearstory / Trimble / Connecteam live — it reads our database.



ARCHITECTURE

How data moves

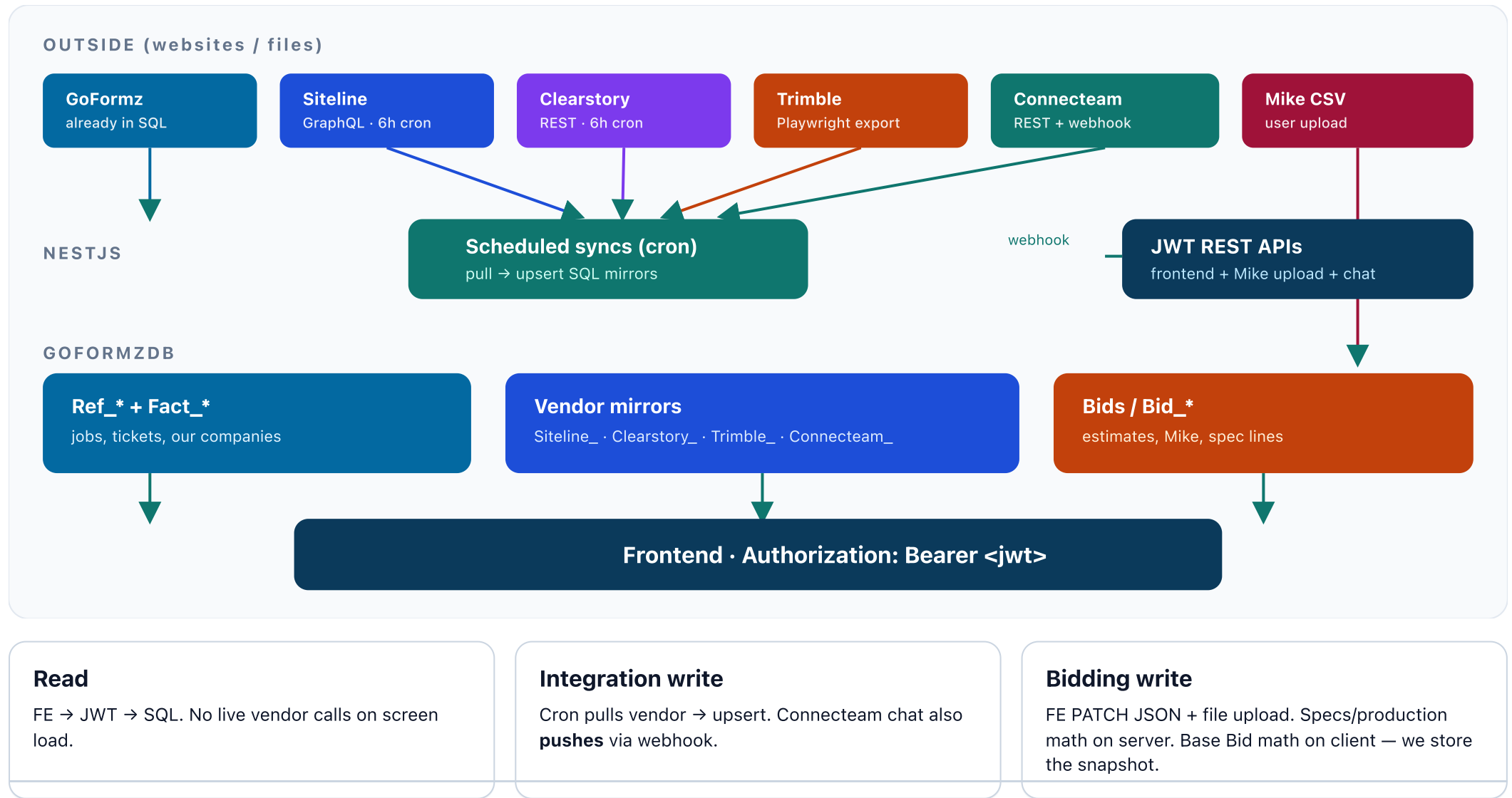


Table prefixes & login

HOW TO READ A TABLE NAME

Ref_*

Shared masters — jobs, our companies

Fact_*

Trucking events — tickets, photos

App_*

Users, roles, settings, files

Siteline_ / Clearstory_ / Trimble_ / Connecteam_

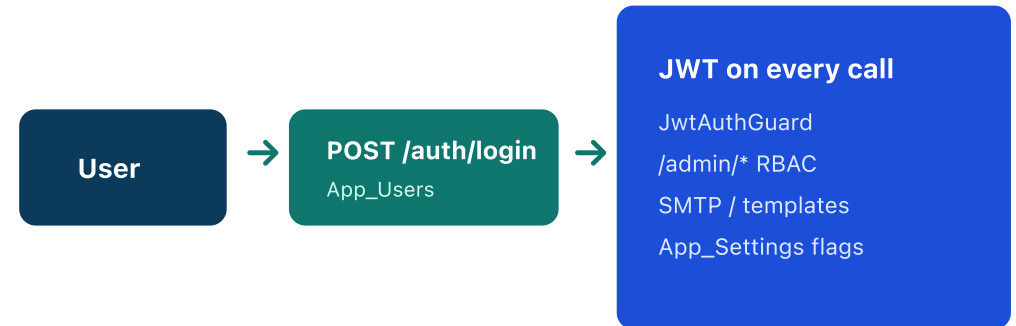
Vendor copies — not CRM

Bid_*

Estimator data with no Ref_* home

Do not invent a second “our companies” or “bidding haulers” table. Bid header FKs:
Bids.OurEntityId → Ref_OurEntities , optional Bids.JobId → Ref_Jobs .

AUTH FLOW



Source is ours: App_Users , App_Roles , App_Permissions . API /auth , /admin/* .

LOOKUPS

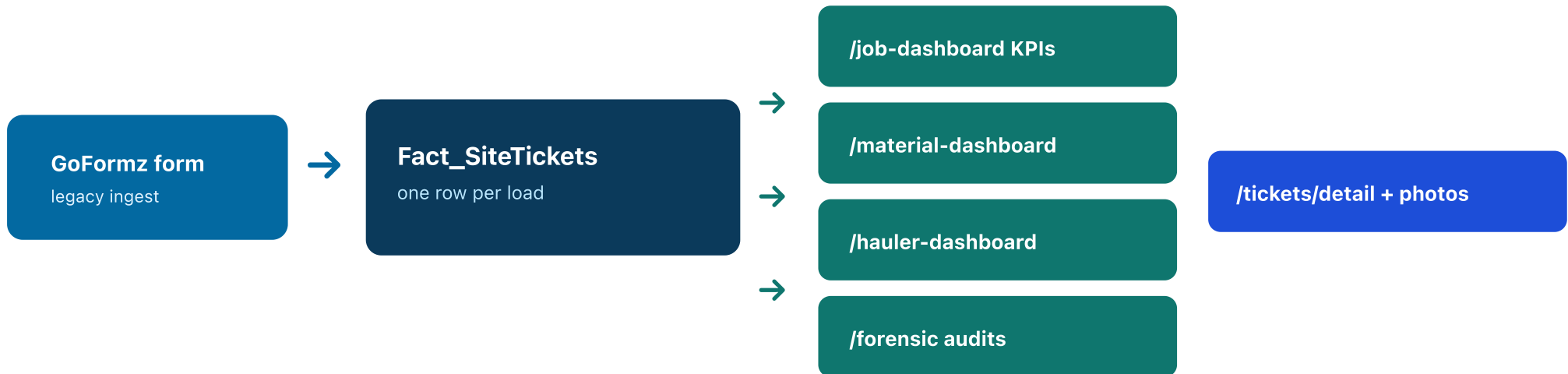
Dashboard dropdowns from Ref_Jobs , Ref_Materials , haulers Ref_ExternalCompanies , sites, Ref_OurEntities → GET /lookups/* . Bidding lists (teams, spec systems...) → GET /lookups/bidding/* — same style, Bid_* tables.

Tickets, jobs, materials, haulers, forensic

What	Original product. Import/export loads, photos, KPIs, late-entry and efficiency audits.
From	GoFormz forms already stored here (<code>GoFormzID</code>). This API does not poll GoFormz.
API	<code>/tickets</code> · <code>/job-dashboard</code> · <code>/material-dashboard</code> · <code>/hauler-dashboard</code> · <code>/forensic</code>

SQL

`Fact_SiteTickets` `Fact_TicketPhotos` `Ref_Jobs` `Ref_Materials`
`Ref_Drivers` `Ref_TruckTypes` `Ref_ExternalCompanies` `Ref_ExternalSites`



Filter with `entityId` (our company) the same way on every dashboard. Not used by bidding.

Four websites → four table families

Same pattern: **cron in, SQL stays, API out**. UI never hits these sites on page load.

Siteline · billing / aging

From	siteline.com GraphQL · one token per GOEL / GOEL DC / DCB
When	~6h cron · SITELINE_*.ENABLED
SQL	Siteline_EntityConfig Siteline_Contracts Siteline_PayApps aging tables
API	/siteline/* + PM / overdue emails

Website → cron → SQL → reports. entityId = Ref_OurEntities via config.

Clearstory · CORs

From	clearstory.build REST
When	~6h cron · CLEARSTORY_SYNC_ENABLED
SQL	Clearstory_Projects Clearstory_Cors customers, tags, rates, Clearstory_ApiPayloads
API	/clearstory/*

Gap alert compares **Siteline SQL vs Clearstory SQL** — no live calls at send time.

Trimble / StructShare · received qty

From	Trimble portal (Playwright — no stable export API)
When	~6h cron · TRIMBLE_SYNC_ENABLED
SQL	Trimble_Projects Trimble_ProjectLineItems raw exports
API	/trimble/*

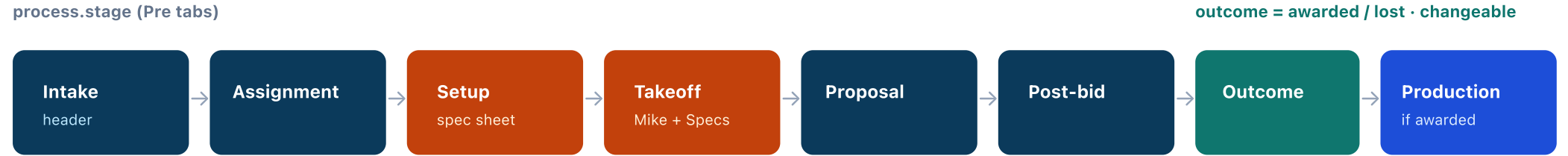
Bidding Specs **Qty Received** = these line items. Bid links
Bids.TrimbleProjectId (often from job number).

Connecteam · hours + chat

From	connecteam.com REST + inbound webhook
When	~6h cron + live webhook for chat
SQL	Connecteam_TimeActivities jobs, users, Connecteam_Messages
API	/connecteam/* · /connecteam/webhooks/inbound

Production report: earned hours (Mike/Trimble) vs **actual** hours here. Match on job number.

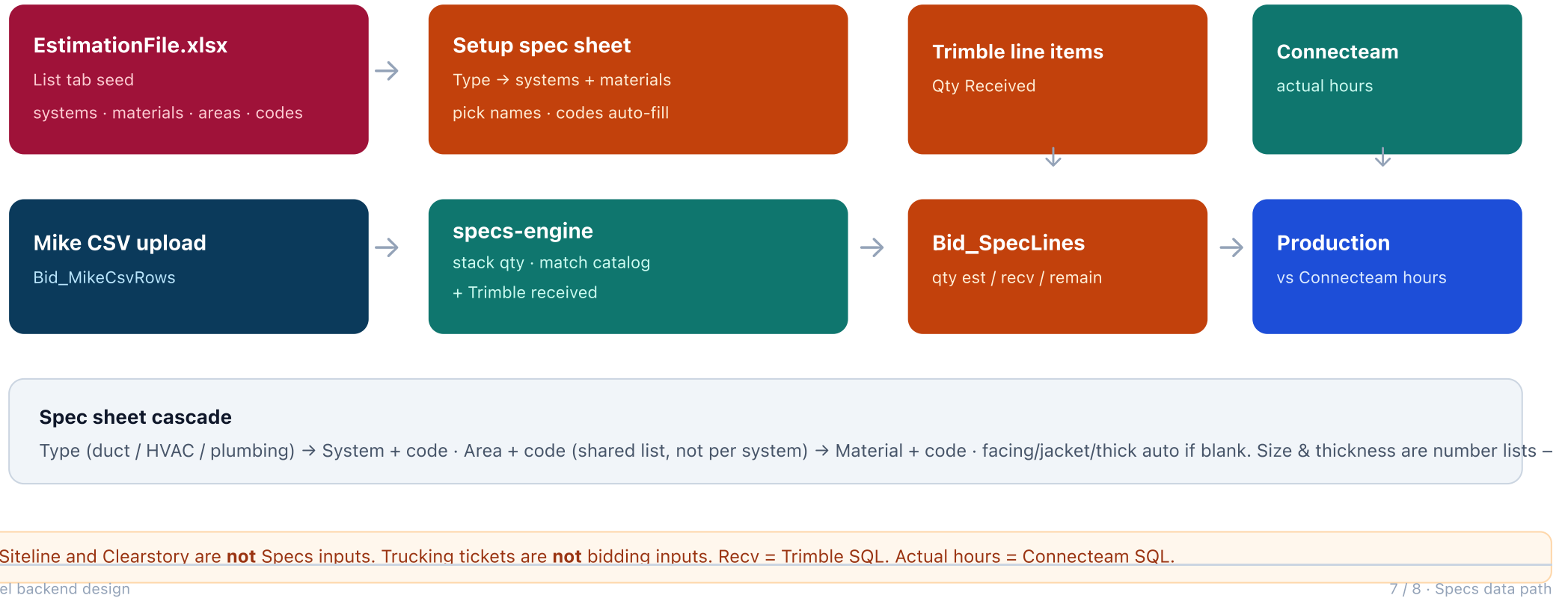
One estimate: invite → takeoff → win/lose → production



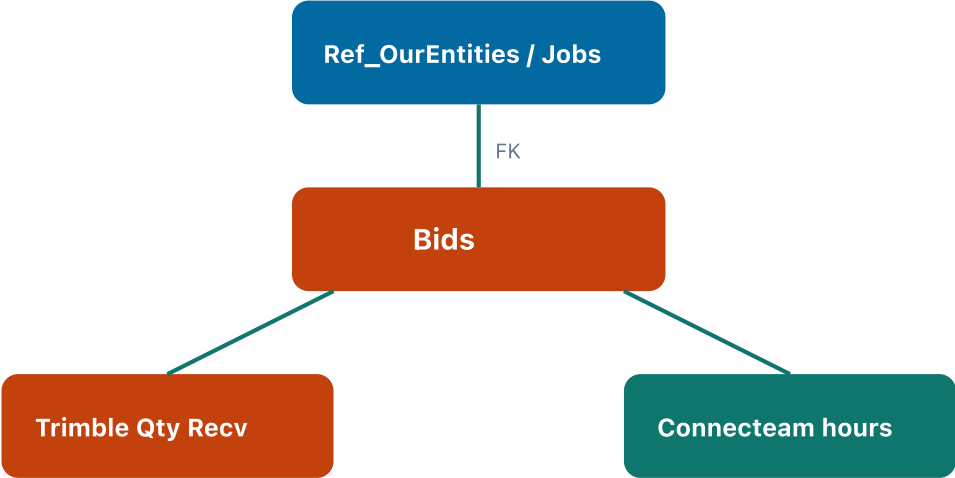
Piece	Comes from	Lives in SQL
Header	User + Ref_OurEntities / Ref_Jobs	Bids
Lifecycle + spec rules	User dropdowns	Bid_Content.ProcessJson
Base Bid / company info	User (Excel-shaped JSON)	Bid_Content JSON columns
Calc snapshot	Frontend Excel engine	Bid_CalcSnapshots
Spec masters + Mike codes	EstimationFile.xlsx List (seed)	Bid_SpecSystems Materials Areas HelperMap
Takeoff qty	Mike CSV upload	Bid_MikeFiles + Bid_MikeCsvRows
Specs grid	Engine: Mike + Trimble recv + catalog	Bid_SpecLines
Production hours	Specs + Connecteam activities	Computed on GET — no extra hours table

Bids.Status draft/submitted/archived locks Estimate math only API: /bids · /lookups/bidding/* · /estimation-files · /production-reports

Setup rules → Mike takeoff → production



Only the joins that actually exist



Siteline ↔ Clearstory

Gap-alert email only. Neither feeds Specs.

Sync switches

All in `.env` . `*_SYNC_ENABLED` . Typical every 6 hours. Crons on ts01 write **production** data.

Three status fields on a bid: **stage** (tab), **outcome** (won/lost), **status** (draft lock). Do not mix them.

WHERE IT LIVES IN THE REPO

Module	Folder
Auth / admin	src/auth · src/admin · src/users
Lookups	src/lookups
Trucking	src/job-dashboard · material · hauler · tickets · forensic
Siteline / Clearstory / Trimble / Connecteam	src/siteline · clearstory · trimble · connecteam
Bidding	src/bidding (+ specs/ · process/)
Entities	src/database/entities

Screen contracts: `docs/FRONTEND_*.md` . This PDF is the map